Assignment 4

Getting started

Implementing a basic simulation

1. Introduce a global clock
2. Perform a step in your simulation in accordance with the global clock.

One of the possible simulation strategy could be -

1. Allocate thread to each populated city
2. Each thread will traverse a path and drop the evacuees

Synchronization

- Population from city A and B arrived in city X - a shelter city at the same time.
- Capacity of X is 100 and is completely empty. Population of A is 200 and population of B is 80.

Assuming that your simulation is handling A and B in a parallel manner
- Threads for both A and B will probe for current capacity of shelter. (100)

Synchronization

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- Capacity of X is 100 and is completely empty. Population of A is 200 and population of B is 80.

Assuming that your simulation is handling A and B in a parallel manner
- Threads for both A and B will probe for current capacity of shelter. (100)
- Since both threads will see 100 empty slots, thread of A will try to drop 100 and thread of B will try to drop 80.
- If both threads succeed, 180 people will be dropped at shelter (exceeds the capacity)

Synchronization

- Only 1 populated city can access the road at a time
- If multiple cities want to use the same road, contention rules must be followed to determine which city can use the road first.
- Some Locking mechanism

Skeleton of simulation

```
Time = 0
while True:
        status = current_status
        if( some conditions)
                Perform step
        Increment Time
```

Perform step will handle all the synchronization + decision making

Decision making - Which path to take ? Should we drop ? …….

10 minutes !!!

- A valid solution must be dumped in the output files before 10 mins
- Make sure to forcibly end the simulation at some point and drop any remaining evacuees in whichever city they are currently present.
- Ensure that your entire output is dumped in the files before 10 mins.